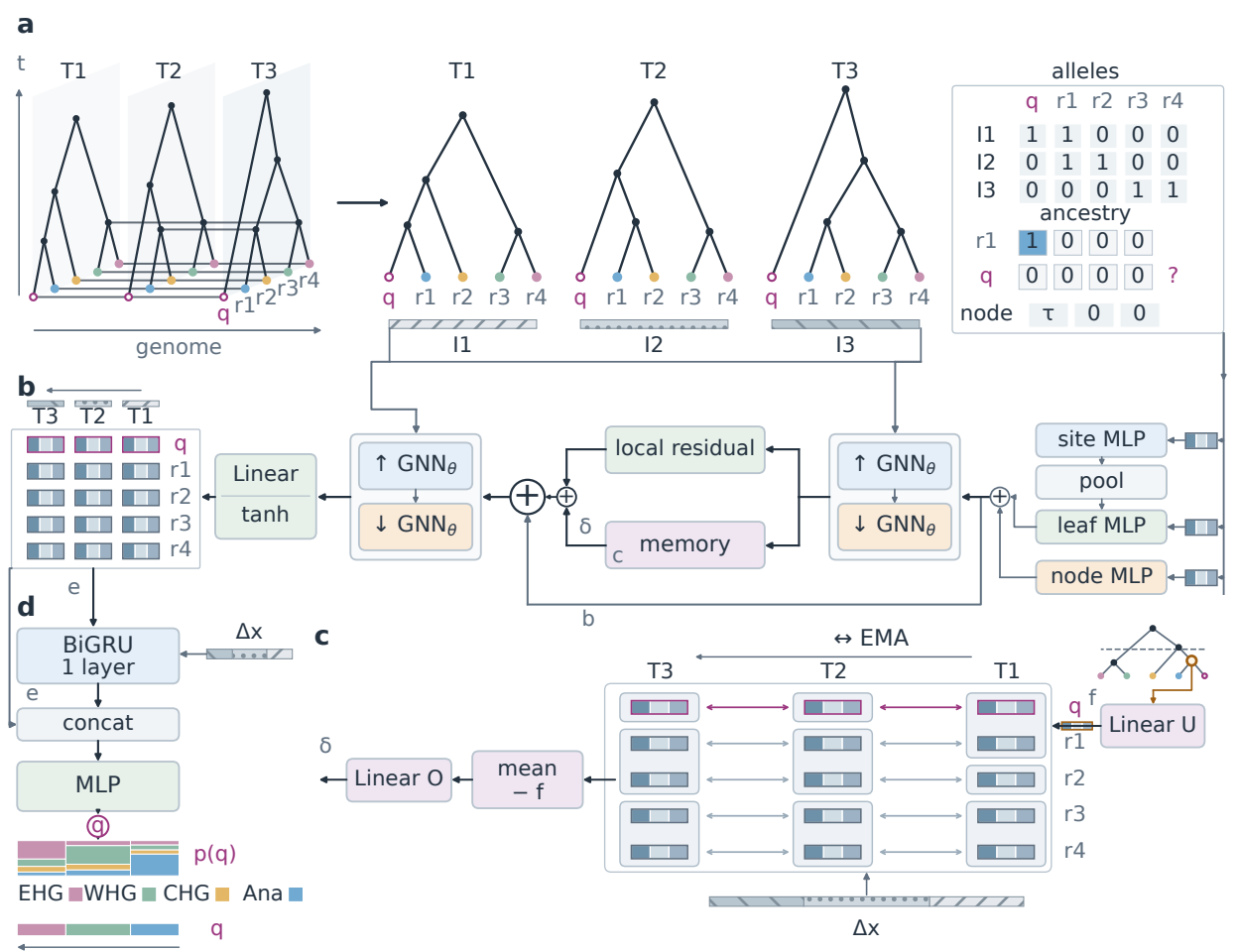

a
t
T1
T2
T3
genome
q
r1
r2
r3
r4
I1
I2
I3
alleles
ancestry
node
τ
b
Linear
tanh
↑ GNN$_\theta$
↓ GNN$_\theta$
local residual
memory
δ
c
site MLP
pool
leaf MLP
node MLP
e
d
BiGRU
1 layer
Δx
concat
MLP
p(q)
EHG
WHG
CHG
Ana
c
↔ EMA
Linear U
Linear O
mean
− f
f